

1 Problem 1: DNS and HTTP Delay

Problem Statement

A user clicks on a link in a Web browser. The IP address corresponding to the URL is not available in the local cache, so a DNS lookup is required.

Suppose that n DNS servers are contacted before the host obtains the IP address. The successive DNS queries incur round-trip times

$$RTT_1, RTT_2, \dots, RTT_n.$$

Let RTT_0 be the round-trip time between the local host and the Web server.

The requested Web page contains exactly one object consisting of a small amount of HTML text. Assume that non-persistent HTTP is used and that the object's transmission time is negligible.

Determine the total time from clicking the link until the object is received.

Solution

There are two main components in the total delay:

1. DNS lookup delay.
2. HTTP delay for establishing a TCP connection and requesting the Web object.

Step 1: DNS Lookup Delay

The browser does not have the IP address in its local cache.

Therefore, DNS queries must be performed.

The problem states that n DNS servers are contacted and the corresponding round-trip times are

$$RTT_1, RTT_2, \dots, RTT_n.$$

Therefore, the total DNS lookup time is

$$T_{\text{DNS}} = RTT_1 + RTT_2 + \dots + RTT_n.$$

In summation notation,

$$T_{\text{DNS}} = \sum_{i=1}^n RTT_i.$$

Step 2: Establishing the TCP Connection

Since non-persistent HTTP is being used, a new TCP connection must be established before the HTTP request is sent.

The TCP connection establishment requires one round-trip time:

$$T_{\text{TCP}} = RTT_0.$$

Step 3: HTTP Request and Response

After the TCP connection has been established, the browser sends the HTTP request.

The server then sends the requested object.

Since the transmission time of the object is negligible, the time required for the request and response is approximately one additional round-trip time:

$$T_{\text{HTTP}} = RTT_0.$$

Step 4: Total Delay

Therefore,

$$T_{\text{total}} = T_{\text{DNS}} + T_{\text{TCP}} + T_{\text{HTTP}}.$$

Substituting the expressions,

$$T_{\text{total}} = \sum_{i=1}^n RTT_i + 2RTT_0$$

Hence, the total time from clicking the link until the Web object is received is

$$RTT_1 + RTT_2 + \cdots + RTT_n + 2RTT_0.$$

2 Problem 2: HTTP Response Time

Problem Statement

Consider the scenario in Problem 1. Suppose that the HTML file references eight additional very small objects located on the same server.

Assume:

- Transmission times of all objects are negligible.
- A new TCP connection must be established before the first HTTP request is sent.
- All DNS lookups have already been completed.
- In the parallel case, up to six objects can be downloaded simultaneously.

Determine the elapsed time for retrieving the base HTML file and all eight referenced objects for:

1. Non-persistent HTTP with no parallel TCP connections.
2. Non-persistent HTTP with six parallel TCP connections.
3. Persistent HTTP without pipelining.

Solution

There are a total of

$$1 + 8 = 9$$

objects:

- 1 base HTML object.
- 8 referenced objects.

Since DNS lookup has already been completed, DNS delay is not included.

2.1 Case 1: Non-Persistent HTTP Without Parallel Connections

For every object under non-persistent HTTP:

1. A new TCP connection is established: $1RTT_0$.
2. HTTP request and response require another $1RTT_0$.

Thus, each object requires

$$2RTT_0.$$

There are nine objects, so

$$T_1 = 9(2RTT_0).$$

Therefore,

$$\boxed{T_1 = 18RTT_0}$$

Alternative way of viewing the calculation

For the base HTML file:

$$2RTT_0.$$

After receiving the HTML file, eight additional objects must be retrieved:

$$8(2RTT_0) = 16RTT_0.$$

Hence,

$$T_1 = 2RTT_0 + 16RTT_0 = 18RTT_0.$$

$$\boxed{T_1 = 18RTT_0}$$

2.2 Case 2: Non-Persistent HTTP with Six Parallel TCP Connections

First, the base HTML file must be retrieved.

Therefore,

$$T_{\text{base}} = 2RTT_0.$$

After receiving the base HTML file, the browser discovers the eight additional objects.

At most six objects can be downloaded simultaneously.

Therefore, the eight objects are divided into two groups:

$$6 + 2.$$

For each object, non-persistent HTTP requires

$$2RTT_0.$$

Because the first six objects are downloaded in parallel, their combined elapsed time is still only

$$2RTT_0.$$

The remaining two objects are then downloaded in parallel, requiring

$$2RTT_0.$$

Thus,

$$T_2 = 2RTT_0 + 2RTT_0 + 2RTT_0.$$

Therefore,

$$\boxed{T_2 = 6RTT_0}$$

Explanation

The sequence is:

$$\underbrace{2RTT_0}_{\text{Base HTML}} + \underbrace{2RTT_0}_{\text{First 6 objects}} + \underbrace{2RTT_0}_{\text{Remaining 2 objects}}$$

Hence,

$$\boxed{T_2 = 6RTT_0}.$$

2.3 Case 3: Persistent HTTP Without Pipelining

For persistent HTTP, only one TCP connection is established.

Therefore, the TCP connection establishment requires

$$1RTT_0.$$

The base HTML file then requires one additional RTT for the HTTP request and response:

$$1RTT_0.$$

After the base HTML file has been received, the browser requests the eight additional objects.

Since pipelining is not used, the browser must wait for each object before requesting the next one.

Therefore, each of the eight objects requires

$$1RTT_0.$$

Thus,

$$T_3 = 1RTT_0 + 1RTT_0 + 8RTT_0.$$

Hence,

$$T_3 = 10RTT_0.$$

Therefore,

$$\boxed{T_3 = 10RTT_0}$$

Final Answers

HTTP Configuration	Total Time
Non-persistent, no parallelism	$18RTT_0$
Non-persistent, 6 parallel connections	$6RTT_0$
Persistent, no pipelining	$10RTT_0$

3 Problem 3: Web Caching

Given Information

The institutional network has:

$$R = 15 \text{ Mbps}$$

access-link transmission rate.

Average object size:

$$L = 1,000,000 \text{ bits.}$$

Average request rate:

$$b = 16 \text{ requests/second.}$$

Average Internet delay:

$$T_{\text{Internet}} = 3 \text{ seconds.}$$

The average access delay is modeled as

$$T_{\text{access}} = \frac{\Delta}{1 - \Delta b},$$

where Δ is the average time required to transmit one object over the access link.

For the cache case, the cache hit delay is assumed to be negligible.

3.1 Part 1: No Cache

Step 1: Calculate Δ

The object size is

$$L = 1,000,000 \text{ bits}$$

and the access-link rate is

$$R = 15 \times 10^6 \text{ bits/sec.}$$

Therefore,

$$\Delta = \frac{L}{R}.$$

Substituting,

$$\Delta = \frac{1,000,000}{15,000,000}.$$

Thus,

$$\Delta = \frac{1}{15} \approx 0.06667 \text{ sec.}$$

Therefore,

$$\boxed{\Delta \approx 0.06667 \text{ sec}}$$

Step 2: Calculate the Access-Link Utilization

The utilization is

$$\rho = \Delta b.$$

Therefore,

$$\rho = \frac{1}{15}(16) = \frac{16}{15} \approx 1.0667.$$

Hence,

$$\boxed{\rho > 1}$$

This is important.

The arrival rate of traffic is greater than the service capacity of the access link.

Therefore, the queue is unstable under the given parameters.

Using the given access-delay model,

$$T_{\text{access}} = \frac{\Delta}{1 - \Delta b} = \frac{1/15}{1 - 16/15}.$$

The denominator is

$$1 - \frac{16}{15} = -\frac{1}{15}.$$

Thus, the formula does not yield a physically meaningful finite queueing delay. The correct interpretation is that the average response time grows without bound because the access link is overloaded.

Therefore,

$$\boxed{T_{\text{response}} = \infty \quad (\text{unstable system})}$$

Conclusion for Part 1

Without a cache, the access-link utilization exceeds 100

$$\boxed{\rho = 1.0667 > 1}.$$

Hence the network cannot serve incoming requests as fast as they arrive, and the average queueing/access delay is unbounded.

Therefore,

$$\boxed{\text{Average response time is unbounded (infinite) under the model.}}$$

3.2 Part 2: Cache with Miss Rate 0.4

The cache miss rate is

$$m = 0.4.$$

Therefore, the cache hit rate is

$$h = 1 - m$$

$$h = 1 - 0.4 = 0.6.$$

Thus,

$$\boxed{h = 0.6}$$

or 60

Step 1: Effective Arrival Rate to the Access Link

Only cache misses need to access the Internet.

Therefore,

$$b_{\text{miss}} = mb.$$

Substituting,

$$b_{\text{miss}} = 0.4(16) = 6.4 \text{ requests/sec.}$$

Thus,

$$\boxed{b_{\text{miss}} = 6.4 \text{ requests/sec}}$$

Step 2: New Access-Link Utilization

We have

$$\rho_{\text{cache}} = \Delta b_{\text{miss}}.$$

Therefore,

$$\rho_{\text{cache}} = \frac{1}{15}(6.4) = 0.42667.$$

Hence,

$$\boxed{\rho_{\text{cache}} \approx 0.4267}$$

The access link is now operating below capacity.

Step 3: Access Delay for a Cache Miss

Using

$$T_{\text{access}} = \frac{\Delta}{1 - \Delta b_{\text{miss}}},$$

we get

$$T_{\text{access}} = \frac{1/15}{1 - (1/15)(6.4)}.$$

Calculate the denominator:

$$1 - \frac{6.4}{15} = 1 - 0.42667 = 0.57333.$$

Therefore,

$$T_{\text{access}} = \frac{0.06667}{0.57333}$$

$$T_{\text{access}} \approx 0.11628 \text{ sec.}$$

Thus,

$$T_{\text{access}} \approx 0.1163 \text{ sec}$$

Step 4: Response Time for a Cache Miss

A cache miss experiences both access-link delay and Internet delay.

Therefore,

$$T_{\text{miss}} = T_{\text{access}} + T_{\text{Internet}}.$$

Thus,

$$T_{\text{miss}} = 0.11628 + 3$$

$$T_{\text{miss}} = 3.11628 \text{ sec.}$$

Therefore,

$$T_{\text{miss}} \approx 3.1163 \text{ sec}$$

Step 5: Average Response Time

A cache hit experiences negligible delay:

$$T_{\text{hit}} \approx 0.$$

60

Therefore,

$$T_{\text{average}} = hT_{\text{hit}} + mT_{\text{miss}}.$$

Substituting,

$$T_{\text{average}} = (0.6)(0) + (0.4)(3.11628).$$

Hence,

$$T_{\text{average}} = 1.24651 \text{ sec.}$$

Therefore,

$$T_{\text{average}} \approx 1.247 \text{ seconds}$$

Final Answers

Scenario	Average Response Time
No cache	Unbounded (link utilization > 1)
Cache miss rate = 0.4	≈ 1.247 sec

4 Problem 4: Parallel and Persistent HTTP

Given Information

Link length:

$$10 \text{ m}$$

Transmission rate:

$$R = 150 \text{ bits/sec.}$$

Data packet length:

$$L_D = 100,000 \text{ bits.}$$

Control packet length:

$$L_C = 200 \text{ bits.}$$

Each HTTP object has size

$$100 \text{ Kbits} = 100,000 \text{ bits.}$$

The first object references 10 additional objects.

Therefore, the total number of objects is

$$1 + 10 = 11.$$

For the parallel case,

$$N = 10.$$

Each parallel TCP connection receives an equal share of the link bandwidth.

Persistent HTTP is used without pipelining in the second part.

4.1 Useful Transmission Times

Data Transmission Time

For a 100,000-bit data packet,

$$T_D = \frac{100,000}{150}$$

$$T_D = 666.6667 \text{ sec.}$$

Thus,

$$\boxed{T_D \approx 666.67 \text{ sec}}$$

Control Packet Transmission Time

For a 200-bit control packet,

$$T_C = \frac{200}{150}$$

$$T_C = 1.3333 \text{ sec.}$$

Thus,

$$T_C \approx 1.333 \text{ sec}$$

We neglect propagation delay because the problem does not provide a propagation speed and the transmission delays dominate the calculation.

4.2 Part 1: Non-Persistent HTTP Without Parallel Connections

For every TCP connection, we account for:

1. TCP three-way handshake.
2. HTTP request.
3. Data transmission.
4. ACK/control transmission.

The three-way TCP handshake contains three control packets:

SYN, SYN-ACK, ACK.

Therefore,

$$T_{\text{handshake}} = 3T_C.$$

Thus,

$$T_{\text{handshake}} = 3(1.3333) = 4 \text{ sec.}$$

The HTTP request takes one control-packet transmission:

$$T_{\text{request}} = T_C = 1.3333 \text{ sec.}$$

The object transmission requires

$$T_D = 666.6667 \text{ sec.}$$

The ACK requires

$$T_{\text{ACK}} = T_C = 1.3333 \text{ sec.}$$

Hence, the time required for one object is

$$T_{\text{object}} = 3T_C + T_C + T_D + T_C.$$

Therefore,

$$T_{\text{object}} = 5T_C + T_D.$$

Substituting,

$$T_{\text{object}} = 5(1.3333) + 666.6667.$$

Thus,

$$T_{\text{object}} = 673.3333 \text{ sec.}$$

There are 11 objects.

Therefore,

$$T_{\text{NP}} = 11(673.3333).$$

Hence,

$$T_{\text{NP}} \approx 7406.67 \text{ sec}$$

Converting into minutes,

$$\frac{7406.67}{60} \approx 123.44 \text{ min.}$$

Therefore,

$$T_{\text{NP}} \approx 123.44 \text{ minutes}$$

4.3 Part 1: Non-Persistent HTTP with 10 Parallel TCP Connections

First, the base object must be downloaded using one TCP connection.

Therefore,

$$T_{\text{base}} = 673.3333 \text{ sec.}$$

After the base object is received, there are 10 additional objects.

They are downloaded using

$$N = 10$$

parallel TCP connections.

The total link bandwidth is

$$150 \text{ bits/sec.}$$

Therefore, each connection receives

$$R_{\text{parallel}} = \frac{150}{10} = 15 \text{ bits/sec.}$$

Thus,

$$R_{\text{parallel}} = 15 \text{ bits/sec}$$

The transmission time of one 100,000-bit data object on one parallel connection is

$$T'_D = \frac{100,000}{15} = 6666.6667 \text{ sec.}$$

Each control packet takes

$$T'_C = \frac{200}{15} = 13.3333 \text{ sec.}$$

Therefore, the TCP handshake time for each parallel connection is

$$T'_{\text{handshake}} = 3T'_C = 40 \text{ sec.}$$

For each of the 10 connections,

$$T_{\text{connection}} = 3T'_C + T'_C + T'_D + T'_C.$$

Therefore,

$$T_{\text{connection}} = 5T'_C + T'_D.$$

Substituting,

$$T_{\text{connection}} = 5(13.3333) + 6666.6667.$$

Hence,

$$T_{\text{connection}} = 6733.3333 \text{ sec.}$$

Since all 10 connections operate in parallel, the 10 objects take only one such interval. Therefore,

$$T_{\text{parallel}} = T_{\text{base}} + T_{\text{connection}}.$$

Thus,

$$T_{\text{parallel}} = 673.3333 + 6733.3333.$$

Hence,

$$T_{\text{parallel}} = 7406.6666 \text{ sec}$$

or approximately

$$T_{\text{parallel}} \approx 123.44 \text{ minutes.}$$

Thus, under the stated equal-bandwidth and control-packet assumptions,

$$T_{\text{parallel}} \approx T_{\text{non-parallel}}$$

Both are approximately

$$7406.67 \text{ sec.}$$

4.4 Part 2: Persistent HTTP Without Pipelining

With persistent HTTP, only one TCP connection is established.

Therefore, there is only one TCP three-way handshake:

$$T_{\text{handshake}} = 3T_C = 4 \text{ sec.}$$

There are 11 objects in total.

Each object requires:

- one HTTP request: T_C ,
- one data transmission: T_D ,
- one ACK/control transmission: T_C .

Therefore,

$$T_{\text{object,persistent}} = T_C + T_D + T_C.$$

Hence,

$$T_{\text{object,persistent}} = 2T_C + T_D.$$

Substituting,

$$T_{\text{object,persistent}} = 2(1.3333) + 666.6667.$$

Thus,

$$T_{\text{object,persistent}} = 669.3333 \text{ sec.}$$

For 11 objects,

$$T_{\text{persistent}} = 3T_C + 11(2T_C + T_D).$$

Substituting,

$$T_{\text{persistent}} = 4 + 11(669.3333).$$

Therefore,

$$T_{\text{persistent}} = 7366.67 \text{ sec}$$

Converting to minutes,

$$\frac{7366.67}{60} \approx 122.78 \text{ minutes.}$$

Therefore,

$$T_{\text{persistent}} \approx 122.78 \text{ minutes}$$

Comparison

The calculated times are:

Method	Download Time
Non-persistent, no parallelism	7406.67 sec
Non-persistent, 10 parallel connections	7406.67 sec
Persistent, no pipelining	7366.67 sec

Under the assumptions stated in the problem, persistent HTTP without pipelining has a slightly smaller calculated time because it avoids repeating the TCP three-way handshake for every object.

$$T_{\text{persistent}} < T_{\text{non-persistent}}$$

5 Problem 5: HTTP/2 Frame Interleaving

Given Information

The Web page contains:

- One video clip.
- Five images.

The video consists of

2000 frames.

Each image consists of

3 frames.

There are five images, so the total number of image frames is

$$5 \times 3 = 15 \text{ frames.}$$

Each frame requires one frame transmission time.

5.1 Part 1: Video Frames Transmitted First

The transmission order is:

$$\underbrace{V_1, V_2, \dots, V_{2000}}_{\text{all video frames}}$$

followed by the image frames.

The question asks for the number of frame transmission times required before all five images have been completely transmitted.

Since all 2000 video frames are transmitted first, we must first spend

$$2000$$

frame transmission times.

Then all 15 image frames must be transmitted:

$$5 \times 3 = 15.$$

Therefore,

$$T_1 = 2000 + 15.$$

Hence,

$T_1 = 2015 \text{ frame transmission times}$

5.2 Part 2: Round-Robin Interleaving

The transmission begins in the following order:

$$V, I_1, I_2, I_3, I_4, I_5.$$

One frame from each available object is transmitted in each round.

Thus, the first round is:

$$V_1, I_{1,1}, I_{2,1}, I_{3,1}, I_{4,1}, I_{5,1}.$$

After the first round, each image has received one frame.

The second round is:

$$V_2, I_{1,2}, I_{2,2}, I_{3,2}, I_{4,2}, I_{5,2}.$$

After the second round, each image has received two frames.

The third round is:

$$V_3, I_{1,3}, I_{2,3}, I_{3,3}, I_{4,3}, I_{5,3}.$$

After these three rounds, each image has received

3 frames.

Since each image contains exactly three frames, all five images are now completely transmitted.

Each round contains six frames:

$$1 \text{ video frame} + 5 \text{ image frames} = 6.$$

There are three rounds.

Therefore,

$$T_2 = 3 \times 6.$$

Hence,

$$T_2 = 18 \text{ frame transmission times}$$

Comparison

Transmission Strategy	Time Until All Images Complete
All video first	2015
Round-robin interleaving	18

The round-robin strategy allows the image frames to be transmitted much earlier instead of waiting for all 2000 video frames.

6 Problem 6: HTTP/2 Prioritization

Given Information

The Web page is the same as in Problem 5.

There are five images:

$$I_1, I_2, I_3, I_4, I_5$$

and one video clip.

Each image consists of three frames:

$$3 \text{ frames/image.}$$

The priorities are:

$$I_1 > I_2 > I_3 > I_4 > I_5 > V.$$

The problem states that a higher-priority object must be completely transmitted before transmission of a lower-priority object begins.

The question asks for the number of frame transmission times required until the second image is completely transmitted.

Solution

Because of strict prioritization, the first image must be completely transmitted before the second image can begin.

Step 1: Transmit Image 1

Image 1 contains

$$3 \text{ frames.}$$

Therefore, transmitting Image 1 requires

$$T_{I_1} = 3$$

frame transmission times.

Step 2: Transmit Image 2

Only after Image 1 has been completely transmitted can Image 2 begin.

Image 2 also contains

$$3 \text{ frames.}$$

Therefore,

$$T_{I_2} = 3.$$

Step 3: Total Time

The total time until Image 2 is completely transmitted is

$$T = T_{I_1} + T_{I_2}.$$

Therefore,

$$T = 3 + 3.$$

Hence,

$T = 6$ frame transmission times

No video frames, Image 3 frames, Image 4 frames, or Image 5 frames need to be transmitted before Image 2 is completed because all of them have lower priority than Image 2.